

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 8 — Reaction Kinetics

Topical Past Paper Question Pack & Comprehensive Mark Scheme

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
TOPIC COVERAGE	8.1 Rate of Reaction · Collision Theory · Tangents · 8.2 Temperature & Boltzmann · 8.3 Catalysis
EXAM SERIES	Past Paper Compilation (2019 – 2025 Paper 2 Theory & Paper 1)
TOTAL MARKS	219 Marks (50 Questions)

HOW TO USE THIS PACK

1. Complete each question carefully in the answer spaces provided, paying close attention to mark allocations.
2. Questions are classified into Easy and Hard tiers to guide progression from core recall to advanced data analysis.
3. Every question includes authentic Cambridge examination codes (e.g. 9701/22/M/J/22) for cross-referencing.
4. Full examiner marking points with point-by-point breakdowns are provided at the end of the document.

SYLLABUS SUBTOPICS COVERED

8.1 Rate of reaction— Rate definitions, collision theory (fruitful collisions, E_a , orientation), experimental monitoring (gas s**8.2 Temperature & Boltzmann**

— Maxwell-Boltzmann distribution of molecular kinetic energies, temperature shifts, activation energ

8.3 Catalysis

— Homogeneous vs heterogeneous catalysts, adsorption-reaction-desorption mechanism, catalys

SUBTOPIC 8.1 — RATE OF REACTION & EXPERIMENTAL METHODS

1 Definition of Reaction Rate & Experimental Methods — 9701/22/M/J/21/Q7(a)–(b)

EASY · Syllabus 8.1

The rate of a chemical reaction is a quantitative measure of how rapidly reactants are converted into products.

(a) Define the term rate of reaction. [2]

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(b) State a suitable experimental method for monitoring the rate of each of the following reactions: [3]

- (i) $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 (ii) $\text{CH}_3\text{COCH}_3(\text{aq}) + \text{I}_2(\text{aq}) \rightarrow \text{CH}_3\text{COCH}_2\text{I}(\text{aq}) + \text{H}^+(\text{aq}) + \text{I}^-(\text{aq})$
 (iii) $\text{Ba}(\text{OH})_2(\text{aq}) + \text{H}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{H}_2\text{O}(\text{l})$

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2 Determining Initial and Instantaneous Rates Using Tangents — 9701/21/O/N/20/Q6(a)–(c)

HARD · Syllabus 8.1

Fig. 8.1 shows a concentration–time graph for the decomposition of a reactant R.

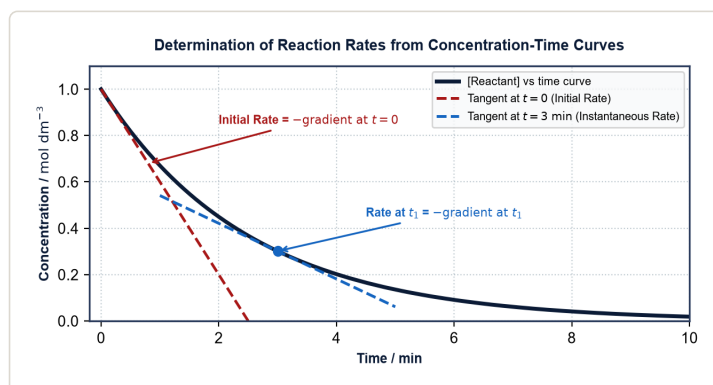


Fig. 8.1: Concentration–time graph showing determination of initial and instantaneous rates using tangents.

(a) Explain how the initial rate of reaction at $t = 0$ is determined from Fig. 8.1. [2]

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(b) Using the tangents shown in Fig. 8.1, calculate the initial rate at $t = 0$ min and the instantaneous rate at $t = 3.0$ min, including units. [3]

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(c) Explain why the reaction rate gradually decreases as time progresses. [2]

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3 Collision Theory & Effective Collisions — 9701/23/M/J/22/Q7(a)–(b)

EASY · Syllabus 8.1

According to collision theory, not all collisions between reactant molecules result in a chemical reaction.

- (a) State two conditions that must be fulfilled during a collision between two molecules for a chemical reaction to occur. [2]

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- (b) Define the term activation energy, E_a . [1]

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4 Effect of Concentration and Pressure on Reaction Rate — 9701/22/F/M/21/Q7(a)–(b)

EASY · Syllabus 8.1

Increasing the concentration of a solution or the pressure of a gaseous mixture increases reaction rate.

- (a) Use collision theory to explain why increasing the concentration of hydrochloric acid increases the rate of reaction with magnesium ribbon. [2]

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- (b) Use collision theory to explain why compressing a gaseous reaction mixture to half its original volume increases the reaction rate. [2]

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5 Effect of Surface Area: Marble Chips vs Powder — 9701/21/M/J/22/Q7(a)–(c)

EASY · Syllabus 8.1

Equal masses of calcium carbonate, CaCO_3 , were reacted with excess 1.0 mol dm^{-3} HCl in two separate experiments:

- Experiment 1: Large marble chips
- Experiment 2: Fine marble powder

- (a) State which experiment has the faster initial rate of reaction. [1]

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- (b) Explain your answer to (a) using collision theory. [2]

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- (c) State whether the total volume of CO_2 gas collected at the end of both reactions will be identical or different. Explain why. [2]

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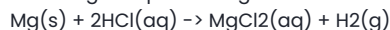
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6 Monitoring Rate via Gas Volume: Magnesium and Acid — 9701/22/O/N/23/Q7(a)–(c)

HARD · Syllabus 8.1

A 0.060 g sample of magnesium ribbon was reacted with 50.0 cm³ of 0.500 mol dm⁻³ HCl:



The volume of H₂ gas collected in a gas syringe was recorded every 15 seconds.

- (a)** Show by calculation which reactant is in excess. (Ar: Mg = 24.3) **[2]**

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- (b)** Calculate the theoretical maximum volume of H₂ gas collected at RTP (molar volume = 24000 cm³ mol⁻¹). **[2]**

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- (c)** Sketch the shape of the graph of volume of H₂ vs time and explain why the gradient eventually reaches zero. **[2]**

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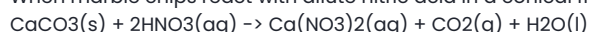
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7 Monitoring Rate via Mass Loss on Digital Balance — 9701/21/M/J/23/Q7(a)–(c)

HARD · Syllabus 8.1

When marble chips react with dilute nitric acid in a conical flask on a digital balance, the mass decreases over time:



- (a)** Explain why the mass of the flask and contents decreases as the reaction proceeds. **[1]**

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- (b)** Explain why a loose cotton wool plug is placed in the neck of the flask during this experiment. **[2]**

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- (c)** State why this mass-loss method is unsuitable for monitoring the reaction between magnesium and hydrochloric acid. **[1]**

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8 Colorimetric Rate Monitoring: Propanone Bromination — 9701/22/M/J/20/Q7(a)–(c)

HARD · Syllabus 8.1

The acid-catalysed reaction between propanone and bromine is:



Bromine is orange-brown, whereas all other reactants and products are colourless.

- (a)** Describe how a colorimeter can be used to monitor the rate of this reaction continuously. **[2]**

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- (b)** State the colour of the filter that should be chosen for the colorimeter in this experiment. **[1]**

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- (c) The absorbance of bromine decreases at a constant rate until all bromine is consumed. State what this reveals about the order of reaction with respect to bromine. [1]

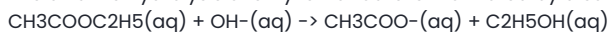
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9 Sampling, Quenching and Titration: Ester Saponification — 9701/21/O/N/22/Q7(a)–(c)

HARD · Syllabus 8.1

The alkaline hydrolysis of ethyl ethanoate is monitored by a sampling technique:



Samples are withdrawn from the reaction mixture at 5-minute intervals.

- (a) Explain what is meant by quenching a reaction sample. [2]

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- (b) Suggest a practical method to quench each sample withdrawn from this hydrolysis reaction. [2]

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- (c) Name the titrant used to determine the concentration of unreacted OH^- in the quenched sample. [1]

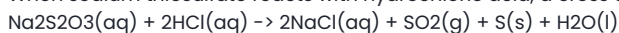
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10 Disappearing Cross Experiment: Sodium Thiosulfate and Acid — 9701/23/O/N/21/Q6(a)–(c)

EASY · Syllabus 8.1

When sodium thiosulfate reacts with hydrochloric acid, a cross drawn on paper beneath the flask disappears:



- (a) Explain why the cross becomes obscured and eventually disappears from view. [1]

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- (b) Explain why $1/\text{time}$ ($1/t$) can be used as a measure of the initial rate of reaction. [2]

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- (c) State one hazard associated with this experiment and a suitable precaution. [2]

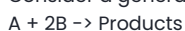
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11 Rate Equations & Orders of Reaction Qualitative Concepts — 9701/22/F/M/23/Q7(a)–(b)

HARD · Syllabus 8.1

Consider a general reaction between reactants A and B:



When [A] is doubled keeping [B] constant, the initial rate doubles.

When [B] is doubled keeping [A] constant, the initial rate quadruples.

- (a) Deduce the order of reaction with respect to A and with respect to B. [2]

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- (b) Write the rate equation for this reaction and state the overall order of reaction. [2]

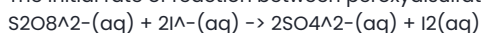
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12 Initial Rates Method Data Analysis — 9701/21/M/J/21/Q8(a)–(b)

HARD · Syllabus 8.1

The initial rate of reaction between peroxydisulfate ions and iodide ions was measured at 25 °C:



- Experiment 1: $[\text{S}_2\text{O}_8^{2-}] = 0.040$, $[\text{I}^{-}] = 0.020$, Rate = $1.6 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Experiment 2: $[\text{S}_2\text{O}_8^{2-}] = 0.080$, $[\text{I}^{-}] = 0.020$, Rate = $3.2 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$
- Experiment 3: $[\text{S}_2\text{O}_8^{2-}] = 0.040$, $[\text{I}^{-}] = 0.040$, Rate = $3.2 \times 10^{-5} \text{ mol dm}^{-3} \text{ s}^{-1}$

- (a) Deduce the order of reaction with respect to $\text{S}_2\text{O}_8^{2-}$ and with respect to I^{-} . [2]

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- (b) Calculate the value of the rate constant, k , and deduce its units. [3]

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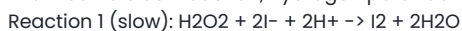
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13 Iodine Clock Reaction Mechanism — 9701/22/M/J/22/Q7(a)–(b)

HARD · Syllabus 8.1

In an iodine clock reaction, hydrogen peroxide reacts with iodide in the presence of a small, fixed amount of thiosulfate and starch:



After all $\text{S}_2\text{O}_3^{2-}$ is consumed, free I_2 reacts with starch to turn suddenly blue-black.

- (a) Explain the role of the thiosulfate ions in this clock reaction. [2]

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- (b) Explain why the blue-black colour appears suddenly rather than gradually developing. [2]

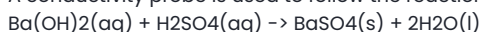
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14 Electrical Conductivity Monitoring: Precipitation Reaction — 9701/22/O/N/21/Q8(a)–(b)

HARD · Syllabus 8.1

A conductivity probe is used to follow the reaction between aqueous barium hydroxide and dilute sulfuric acid:



- (a) Explain why the electrical conductivity drops to near zero at the exact equivalence point. [2]

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- (b) Explain what happens to the electrical conductivity if excess sulfuric acid is subsequently added. [2]

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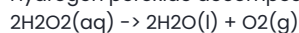
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15 Decomposition of Hydrogen Peroxide: MnO₂ Catalysis — 9701/21/O/N/23/Q7(a)–(c)

EASY · Syllabus 8.1

Hydrogen peroxide decomposes slowly at room temperature:



Adding a small spatula of black manganese(IV) oxide powder, MnO₂, causes vigorous effervescence.

- (a) State the role of MnO₂ in this reaction. [1]

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- (b) Describe an experiment to demonstrate that MnO₂ is unchanged chemically and in mass at the end of the reaction. [3]

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16 Calculating Initial Rate from Gas Evolution Curve — 9701/23/M/J/20/Q6(a)–(b)

EASY · Syllabus 8.1

In an experiment reacting zinc granules with hydrochloric acid, 48.0 cm³ of H₂ was collected in the first 60 seconds. A tangent drawn to the curve at t = 0 passes through (0, 0) and (40 s, 44.0 cm³).

- (a) Calculate the initial rate of reaction in cm³ s⁻¹. [2]

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- (b) Calculate the average rate of reaction over the first 60 seconds and explain why it is lower than the initial rate. [2]

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17 Collision Frequency vs Fraction of Successful Collisions — 9701/22/F/M/24/Q6(a)–(b)

HARD · Syllabus 8.1

Students often confuse the effects of changing concentration with the effects of changing temperature.

- (a) State whether increasing the concentration of a reactant increases the collision frequency, the fraction of collisions with E ≥ E_a, or both. [1]

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- (b) State whether increasing the temperature of a reaction mixture increases the collision frequency, the fraction of collisions with E ≥ E_a, or both, and identify which factor is primarily responsible for the rate increase. [3]

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18 Decay of Reaction Rate with Reactant Consumption — 9701/21/M/J/24/Q8(a)–(b)

EASY · Syllabus 8.1

During the reaction $A \rightarrow B$, the instantaneous rate was measured at different times:

- At $t = 0$ min: Rate = $0.080 \text{ mol dm}^{-3} \text{ min}^{-1}$
- At $t = 2$ min: Rate = $0.040 \text{ mol dm}^{-3} \text{ min}^{-1}$
- At $t = 4$ min: Rate = $0.020 \text{ mol dm}^{-3} \text{ min}^{-1}$
- At $t = 6$ min: Rate = $0.010 \text{ mol dm}^{-3} \text{ min}^{-1}$

- (a) State the term used to describe the constant time interval required for the rate to halve. [1]

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- (b) Deduce the order of reaction with respect to A from this data, justifying your answer. [2]

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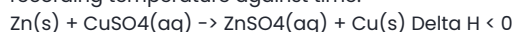
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19 Thermometric Rate Monitoring: Zinc and Copper Sulfate — 9701/22/O/N/24/Q7(b)

HARD · Syllabus 8.1

The rate of the exothermic displacement reaction between zinc and copper(II) sulfate can be followed thermometrically by recording temperature against time:



- (a) Explain how the initial rate of this reaction can be estimated from a temperature-time graph. [2]

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- (b) Explain why this thermometric method becomes inaccurate if heat loss to the surroundings is significant. [2]

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20 Surface Area and Dust Explosions — 9701/21/O/N/24/Q7(a)–(b)

EASY · Syllabus 8.1

A lump of coal burns steadily in air, whereas finely dispersed coal dust in a mine can cause a catastrophic explosion when ignited.

- (a) Explain why coal dust reacts with explosive speed compared to lump coal. [2]

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- (b) State another industrial environment prone to dust explosions and identify the particulate matter. [2]

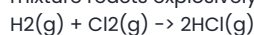
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21 Photochemical Reaction Rates: $\text{H}_2 + \text{Cl}_2$ — 9701/23/M/J/23/Q6(a)–(b)

HARD · Syllabus 8.1

A mixture of hydrogen and chlorine gas stored in the dark reacts imperceptibly slowly. When exposed to bright ultraviolet light, the mixture reacts explosively:



- (a) Explain the role of UV light in initiating this photochemical reaction. [2]

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- (b) Write the initiation step equation showing the formation of reactive chlorine radicals. [1]

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22 Rates of Halogenoalkane Hydrolysis — 9701/22/M/J/25/Q6(a)–(c)

HARD · Syllabus 8.1

Equal amounts of 1-chlorobutane, 1-bromobutane, and 1-iodobutane are warmed with aqueous silver nitrate and ethanol in separate test tubes.

- (a) State the order of reactivity of the three halogenoalkanes from slowest to fastest. [1]

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- (b) Explain this order of reaction rates in terms of carbon–halogen bond enthalpy. [3]

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- (c) State the precipitate colour observed with 1-iodobutane. [1]

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23 Continuous Monitoring vs Sampling Methods — 9701/21/F/M/25/Q7(a)–(b)

EASY · Syllabus 8.1

Kinetic studies can be conducted using continuous measurement or sampling techniques.

- (a) State one major advantage of a continuous monitoring method (such as colorimetry) over a sampling and titration method. [1]

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- (b) State one limitation or source of error specific to the sampling and quenching method. [2]

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24 Units of Reaction Rate and Rate Constants — 9701/22/O/N/25/Q7(a)–(b)

EASY · Syllabus 8.1

Standard SI units and consistent dimensions are crucial in reaction kinetics.

- (a) State the standard SI unit for reaction rate in homogeneous solution. [1]

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- (b) Deduce the units of the rate constant, k , for a second-order rate equation: $\text{Rate} = k[A]^2$. [2]

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25 Reactivity Trends Down Group 1: Kinetics vs Thermodynamics — 9701/23/O/N/24/Q6(a)–(b)

EASY · Syllabus 8.1

When small pieces of sodium and potassium are added to cold water in separate troughs, potassium reacts much more violently and rapidly, igniting with a lilac flame.

- (a) Explain why potassium reacts more rapidly with water than sodium in terms of first ionisation energy. [2]

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- (b) Write the balanced chemical equation with state symbols for the reaction of potassium with water. [2]

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SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS

26 Maxwell-Boltzmann Energy Distribution at Two Temperatures — 9701/22/M/J/21/Q8(a)–(d)

HARD · Syllabus 8.2

Fig. 8.2 shows the Maxwell-Boltzmann distribution of molecular kinetic energies for a gas sample at temperatures T_1 and T_2 (where $T_2 > T_1$).

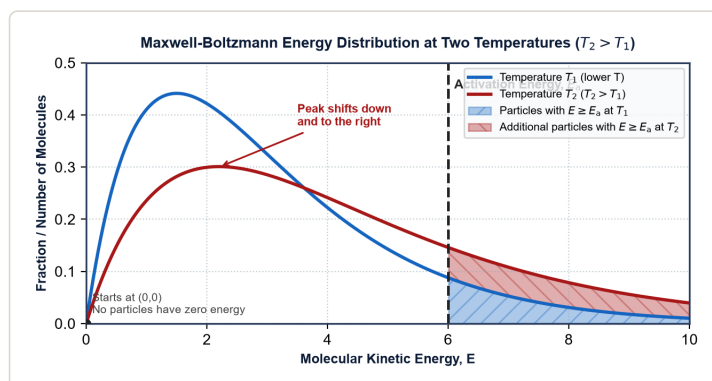


Fig. 8.2: Maxwell-Boltzmann kinetic energy distribution curves at temperatures T_1 and T_2 .

- (a) Explain why the distribution curve starts at the origin (0,0). [1]

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- (b) Describe how the shape of the curve changes when temperature is increased from T_1 to T_2 . [2]

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- (c) State what the total area under each curve represents. [1]

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- (d) With reference to Fig. 8.2, explain why a modest increase in temperature results in a substantial increase in the rate of reaction.

[2]

SUBTOPIC 8.3 — HOMOGENEOUS & HETEROGENEOUS CATALYSIS

27 Effect of a Catalyst on the Maxwell-Boltzmann Distribution — 9701/21/O/N/21/Q7(a)–(c)

HARD · Syllabus 8.3

Fig. 8.3 shows the effect of adding a catalyst to a reaction mixture on the Maxwell-Boltzmann distribution curve.

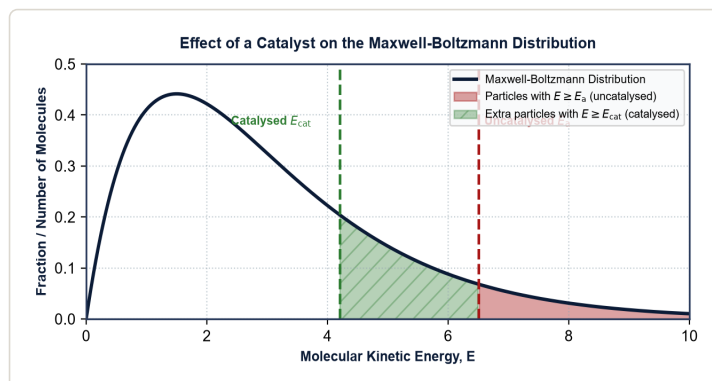


Fig. 8.3: Maxwell-Boltzmann distribution showing shift in activation energy with a catalyst.

- (a) Define the term catalyst.
- (b) With reference to Fig. 8.3, explain how adding a catalyst increases the rate of reaction at constant temperature.
- (c) State what happens to the shape of the Maxwell-Boltzmann curve when a catalyst is added at constant temperature.

[2]

[2]

[1]

28 Mechanism of Heterogeneous Catalysis — 9701/23/M/J/22/Q8(a)–(d)

HARD · Syllabus 8.3

Fig. 8.4 illustrates the four sequential stages of a heterogeneous catalytic reaction on a solid metal surface.



Fig. 8.4: Four-stage mechanism of heterogeneous catalysis on a solid active surface.

- (a) Define a heterogeneous catalyst. [1]
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- (b) Describe what occurs during Stage 1 (Adsorption) and Stage 2 (Bond Weakening). [2]
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- (c) Describe what occurs during Stage 3 (Surface Reaction) and Stage 4 (Desorption). [2]
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- (d) Explain why a solid catalyst that bonds too strongly to reactant molecules becomes ineffective. [2]
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SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS

29 The 10 °C Rule of Thumb in Reaction Kinetics — 9701/22/F/M/22/Q6(a)–(b)

EASY · Syllabus 8.2

As a general rule of thumb for reactions with moderate activation energy at room temperature, an increase of 10 °C approximately doubles the reaction rate.

- (a) Explain why an increase of 10 °C causes a collision frequency increase of only about 2%, yet doubles the overall reaction rate. [3]
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- (b) Predict how many times faster a reaction will proceed at 50 °C compared to 20 °C according to this rule of thumb. [2]
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SUBTOPIC 8.3 — HOMOGENEOUS & HETEROGENEOUS CATALYSIS

30 Classification: Homogeneous vs Heterogeneous Catalysts — 9701/21/M/J/22/Q8(a)–(b)

EASY · Syllabus 8.3

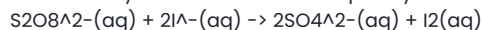
Catalysts are classified according to the physical phase in which they operate relative to the reactants.

- (a) Classify each of the following catalytic processes as either homogeneous or heterogeneous: [4]
- (i) Haber process: Iron catalyst with gaseous N₂ and H₂
- (ii) Atmospheric oxidation: Gaseous NO₂ catalysing reaction of gaseous SO₂ and O₂
- (iii) Esterification: Aqueous H₂SO₄ catalysing liquid ethanoic acid and ethanol
- (iv) Catalytic converter: Platinum–rhodium solid mesh with exhaust gases
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31 Homogeneous Catalysis: Persulfate and Iodide by Fe²⁺/Fe³⁺ — 9701/22/O/N/22/Q6(a)–(c)

HARD · Syllabus 8.3

The uncatalysed reaction between peroxydisulfate ions and iodide ions is very slow at room temperature:



Adding a few drops of aqueous iron(II) or iron(III) ions dramatically accelerates the reaction.

- (a)** Explain why the uncatalysed reaction has an exceptionally high activation energy. **[2]**

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- (b)** Write two sequential chemical equations demonstrating how Fe²⁺(aq) catalyses this reaction. **[2]**

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- (c)** Explain why Fe³⁺(aq) is equally effective as Fe²⁺(aq) as a catalyst for this reaction. **[1]**

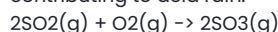
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32 Homogeneous Atmospheric Catalysis: Acid Rain Formation — 9701/23/O/N/22/Q6(a)–(b)

HARD · Syllabus 8.3

In the upper atmosphere, nitrogen oxides act as homogeneous catalysts in the conversion of sulfur dioxide into sulfur trioxide, contributing to acid rain:



- (a)** Write two equations showing how NO₂(g) catalyses the atmospheric oxidation of SO₂. **[2]**

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- (b)** Explain why NO₂ acts as a catalyst rather than a consumed reactant in this environmental mechanism. **[1]**

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33 Automobile Catalytic Converters — 9701/22/F/M/23/Q8(a)–(c)

HARD · Syllabus 8.3

Modern vehicle exhaust systems incorporate three-way catalytic converters containing platinum, palladium, and rhodium supported on a ceramic honeycomb.

- (a)** State the purpose of the ceramic honeycomb structure. **[1]**

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- (b)** Write a balanced equation for the simultaneous catalytic removal of carbon monoxide and nitrogen monoxide. **[2]**

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- (c)** Explain why catalytic converters do not reduce carbon dioxide emissions. **[1]**

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34 Catalyst Poisoning in Industrial Processes — 9701/21/M/J/23/Q8(a)–(b)

EASY · Syllabus 8.3

Catalyst poisoning is the deactivation of a catalyst by chemical substances present in the feed stream.

- (a) Explain at the molecular level how catalyst poisoning occurs in heterogeneous catalysis. [2]

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- (b) Identify a common poison for the iron catalyst in the Haber process and for catalytic converters in cars. [2]

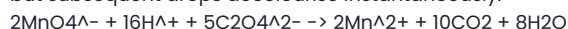
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35 Autocatalysis: Permanganate and Ethanedioate — 9701/22/M/J/23/Q6(a)–(c)

HARD · Syllabus 8.3

When potassium manganate(VII) is titrated against warm acidified ethanedioic acid, the initial purple colour disappears very slowly, but subsequent drops decolourise instantaneously:



- (a) Define the term autocatalysis. [1]

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- (b) Identify the autocatalyst in this reaction. [1]

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- (c) Sketch the shape of the reaction rate vs time graph for an autocatalysed reaction, showing the characteristic induction period. [2]

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36 Biological Catalysts: Enzymes — 9701/21/O/N/23/Q8(a)–(c)

EASY · Syllabus 8.3

Enzymes are complex globular protein molecules that catalyse biochemical processes in living organisms.

- (a) State two differences between enzymes and inorganic catalysts such as platinum. [2]

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- (b) Explain why enzyme-catalysed reactions exhibit a maximum rate at an optimum temperature (typically around 37–40 °C) and drop rapidly to zero above 60 °C. [3]

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37 Comparing Catalysed vs Uncatalysed Reaction Coordinate Profiles — 9701/22/O/N/23/Q8(b)

EASY · Syllabus 8.3

Consider a reversible exothermic reaction: $A + B \rightleftharpoons C + D$.

- (a) Explain the effect of a catalyst on: (i) the enthalpy change ΔH , (ii) the activation energy E_a , and (iii) the activation energy of the reverse reaction. [3]

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- (b) Explain why a catalyst cannot turn a thermodynamically non-spontaneous reaction into a spontaneous reaction. [1]

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SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS

38 Mathematical Features of the Maxwell-Boltzmann Distribution — 9701/23/M/J/24/Q8(a)–(b)

HARD · Syllabus 8.2

The Maxwell-Boltzmann distribution curve possesses precise mathematical characteristics.

- (a) Explain why the curve never touches the energy axis (x-axis) at high energies. [1]

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- (b) Explain why the most probable energy, E_{mp} (the peak of the curve), is not equal to the average / mean kinetic energy, E_{mean} . [2]

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SUBTOPIC 8.3 — HOMOGENEOUS & HETEROGENEOUS CATALYSIS

39 Iron Catalyst in the Haber Process: Role and Preparation — 9701/21/M/J/24/Q9(a)–(b)

EASY · Syllabus 8.3

In the Haber process, finely divided iron mixed with potassium and aluminium oxide promoters is used as a heterogeneous catalyst.

- (a) Explain why the iron catalyst is manufactured in a finely divided, porous form. [2]

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- (b) Explain what would happen to the operational temperature of the Haber process plant if the iron catalyst were removed. [2]

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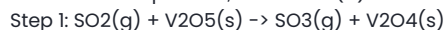
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40 Vanadium(V) Oxide in the Contact Process: Redox Cycle — 9701/22/M/J/24/Q9(a)–(c)

HARD · Syllabus 8.3

In the Contact process, vanadium(V) oxide acts as a catalyst by undergoing a two-step redox cycle:



(a) Deduce the oxidation state of vanadium in V_2O_5 and in V_2O_4 . **[2]**

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(b) Identify the role of SO_2 in Step 1 and the role of O_2 in Step 2 in terms of oxidation and reduction. **[2]**

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(c) Explain why transition metals and their compounds are uniquely effective catalysts. **[2]**

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41 Common Misconceptions: Catalysts and Equilibrium Position — 9701/23/O/N/23/Q7(a)–(b)

EASY · Syllabus 8.3

A student states: 'Adding a catalyst increases the rate of reaction, which shifts the equilibrium to the right to produce a higher yield of products.'

(a) Explain why this student's statement is fundamentally incorrect. **[2]**

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(b) State the exact effect of a catalyst on: (i) the rate of forward reaction, (ii) the rate of reverse reaction, and (iii) the equilibrium constant K_c . **[3]**

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42 Adsorption Energetics: Physisorption vs Chemisorption — 9701/22/F/M/25/Q8(a)–(b)

HARD · Syllabus 8.3

Adsorption onto a solid catalyst can occur via physisorption or chemisorption.

(a) Distinguish between physisorption and chemisorption in terms of the types of forces involved and enthalpy change. **[2]**

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(b) Explain why chemisorption is essential for catalysis to take place. **[2]**

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43 Acid-Catalysed Ester Hydrolysis: Homogeneous Mechanism — 9701/21/F/M/25/Q8(a)–(b)**HARD · Syllabus 8.3**

The hydrolysis of ethyl ethanoate is catalysed by dilute hydrochloric acid:
 $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \xrightarrow{[\text{H}^+]} \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$

- (a) Explain why $\text{H}^+(\text{aq})$ acts as a homogeneous catalyst in this aqueous solution. [1]

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- (b) Explain how protonation of the carbonyl oxygen atom in the ester accelerates nucleophilic attack by water. [2]

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44 Transition Metal d-Orbitals in Catalysis — 9701/22/M/J/25/Q8(a)–(b)**HARD · Syllabus 8.3**

Transition elements and their compounds make up the vast majority of industrial catalysts.

- (a) Explain how the presence of partially filled 3d subshells enables transition metals to act as catalysts. [2]

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- (b) Contrast the catalytic activity of transition metals (e.g. Fe, Ni) with s-block metals (e.g. Na, Ca). [2]

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45 Inert Gas Addition and Reaction Kinetics — 9701/23/M/J/25/Q7(a)–(b)**EASY · Syllabus 8.3**

Consider the gaseous reaction: $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$.

- (a) State and explain the effect on the rate of reaction when argon gas is added at constant volume. [2]

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- (b) State and explain the effect on the rate of reaction when argon gas is added at constant total pressure. [2]

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46 Heterogeneous Catalytic Hydrogenation of Alkenes — 9701/21/O/N/25/Q8(a)–(c)**HARD · Syllabus 8.3**

Vegetable oils are hardened into margarine by catalytic hydrogenation using a nickel catalyst at 150°C :
 $\text{R}-\text{CH}=\text{CH}-\text{R}' + \text{H}_2 \rightarrow \text{R}-\text{CH}_2-\text{CH}_2-\text{R}'$

- (a) State the type of catalysis involved. [1]

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- (b) Describe how H₂ molecules interact with the nickel surface during the reaction. [2]

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- (c) Explain why trans-fats can be formed as undesirable by-products during partial hydrogenation. [2]

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SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS**47 Temperature Coefficient Q₁₀ & Arrhenius Concept — 9701/22/O/N/25/Q8(a)-(b)**

HARD · Syllabus 8.2

The temperature coefficient Q₁₀ is defined as the ratio of reaction rates at temperatures separated by 10 °C:

$$Q_{10} = \text{Rate}(T + 10) / \text{Rate}(T)$$

- (a) Explain why reactions with very high activation energies have a larger Q₁₀ value than reactions with low activation energies. [2]

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- (b) State what happens to the value of the rate constant, k, as temperature increases. [1]

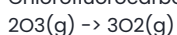
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SUBTOPIC 8.3 — HOMOGENEOUS & HETEROGENEOUS CATALYSIS**48 Homogeneous Ozone Depletion by Chlorine Radicals — 9701/23/O/N/25/Q7(a)-(c)**

HARD · Syllabus 8.3

Chlorofluorocarbons (CFCs) release chlorine radicals in the stratosphere, catalysing ozone breakdown:



- (a) Write the two propagation steps for this catalytic destruction cycle. [2]

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- (b) Show that adding the two propagation steps yields the overall stoichiometric equation. [1]

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- (c) Explain why a single chlorine radical can destroy up to 100,000 ozone molecules. [1]

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49 Engineering Design of Heterogeneous Catalytic Converters — 9701/21/M/J/25/Q8(a)-(b)

EASY · Syllabus 8.3

Automobile catalytic converters use a ceramic monolith structure with thousands of micro-channels.

- (a) State two engineering advantages of using a micro-channel honeycomb over packed catalyst pellets. [2]

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- (b) Explain why cold vehicle exhaust emits significantly higher pollutants during the first two minutes after starting the engine. [2]

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SUBTOPIC 8.2 — TEMPERATURE & MAXWELL-BOLTZMANN DISTRIBUTIONS

50 Comprehensive Synoptic: Kinetics, Boltzmann Distributions & Catalyst Surface Chemistry — 9701/22/O/N/25/Q8(a)-(c) [2] Syllabus 8.2

In the industrial synthesis of ammonia, $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$, the uncatalysed reaction has an activation energy of +335 kJ mol⁻¹, whereas the iron-catalysed reaction has an activation energy of +160 kJ mol⁻¹.

- (a) Explain why the uncatalysed reaction has such an immense activation energy (+335 kJ mol⁻¹). [2]

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- (b) Describe how the iron catalyst lowers this activation energy to +160 kJ mol⁻¹ by detailing the surface adsorption process. [2]

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- (c) Draw a labelled Maxwell-Boltzmann distribution sketch showing both uncatalysed E_a (+335) and catalysed E_{cat} (+160). [2]

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- (d) Explain how the iron catalyst enables industrial plants to operate at 450 °C rather than 800 °C, and explain the economic benefit. [2]

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Mark Scheme — Topic 8 — Reaction Kinetics

CAMBRIDGE INTERNATIONAL AS LEVEL CHEMISTRY (9701) — THEORY MARKING POINTS

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
1(a)	The change in concentration of a reactant or product (1); per unit time (1)	2
1(b)	(i) Measuring the volume of CO ₂ gas evolved using a gas syringe / measuring loss of mass on a digital balance (1); (ii) Colorimetry / measuring the disappearance of brown I ₂ color over time (1); (iii) Electrical conductivity / measuring the decrease in conductivity as ions precipitate (1)	3
2(a)	Draw a tangent to the curve at time $t = 0$ (1); Calculate the gradient of this tangent: initial rate = -gradient (1)	2
2(b)	Initial rate ($t = 0$): gradient = $(1.0 - 0) / (2.5 - 0) = 0.40 \text{ mol dm}^{-3} \text{ min}^{-1}$ (1); Instantaneous rate ($t = 3 \text{ min}$): gradient = $0.12 \text{ mol dm}^{-3} \text{ min}^{-1}$ (1); Units: $\text{mol dm}^{-3} \text{ min}^{-1}$ (1)	3
2(c)	As the reaction proceeds, reactant particles are consumed and their concentration decreases (1); The frequency of collisions between reactant particles per unit time decreases (1)	2
3(a)	The colliding particles must possess kinetic energy greater than or equal to the activation energy ($E > E_a$) (1); The particles must collide with the correct molecular orientation / collision geometry (1)	2
3(b)	The minimum kinetic energy required by colliding particles to initiate a chemical reaction (1)	1
4(a)	More reactant particles are present per unit volume (1); This increases the frequency of collisions between H ⁺ ions and magnesium per second, leading to more successful collisions per unit time (1)	2
4(b)	Halving the volume doubles the concentration / number of gas particles per unit volume (1); This increases the frequency of collisions between reacting gas molecules per unit time (1)	2
5(a)	Experiment 2 (fine marble powder) (1)	1
5(b)	Fine powder has a much greater surface area exposed to the acid per unit mass (1); More calcium carbonate particles are available for collisions, increasing the collision frequency per unit time with H ⁺ ions (1)	2
5(c)	Identical total volume of CO ₂ (1); Both experiments used equal masses of CaCO ₃ and excess acid, so the limiting reactant amount is identical (1)	2
6(a)	$n(\text{Mg}) = 0.060 / 24.3 = 0.00247 \text{ mol}$; $n(\text{HCl}) = 0.500 \times 0.0500 = 0.0250 \text{ mol}$ (1); Reaction requires $2 \times 0.00247 = 0.00494 \text{ mol}$ HCl; since 0.0250 mol is present, HCl is in large excess (1)	2
6(b)	$n(\text{H}_2) = n(\text{Mg}) = 0.00247 \text{ mol}$; Volume = $0.00247 \times 24000 = 59.3 \text{ cm}^3$ (allow $59.0 - 59.5 \text{ cm}^3$) (2)	2
6(c)	Steep curve starting at origin, plateauing horizontally at 59.3 cm^3 (1); Gradient becomes zero because all magnesium has reacted completely (reaction has stopped) (1)	2
7(a)	Carbon dioxide gas, CO ₂ , escapes from the open flask into the atmosphere (1)	1
7(b)	The cotton wool plug allows CO ₂ gas to escape freely (1); while preventing acid spray / liquid droplets from splashing out of the flask, which would cause an erroneous mass loss (1)	2
7(c)	Hydrogen gas has an extremely low relative molecular mass ($M_r = 2.0$); the total mass loss would be too small to measure accurately on a standard laboratory balance (1)	1
8(a)	Place the reaction mixture in a cuvette in the colorimeter and record absorbance at regular time intervals (1); Absorbance is directly proportional to [Br ₂], so rate of change of absorbance equals rate of reaction (1)	2
8(b)	Blue filter (complementary colour to orange-brown) (1)	1
8(c)	Zero order with respect to bromine (rate is independent of [Br ₂]) (1)	1
9(a)	Stopping / instantaneously freezing the chemical reaction at a specific time (1); so that the concentrations of reactants remain fixed during analysis (1)	2
9(b)	Rapidly chilling the sample in ice-water / adding a known excess of cold standard acid (1); to neutralise and halt OH ⁻ reaction immediately (1)	2
9(c)	Standard hydrochloric acid (or back-titration with standard NaOH if acid was added to quench) (1)	1

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
10(a)	Formation of insoluble colloidal sulfur precipitate which scatters light and causes yellow turbidity (1)	1
10(b)	The cross disappears when a fixed, constant amount/mass of sulfur precipitate has formed (1); Since rate = amount/time, rate is directly proportional to 1/t (1)	2
10(c)	Hazard: Toxic / choking sulfur dioxide gas (SO ₂) produced (1); Precaution: Perform in a fume cupboard / ensure good ventilation (1)	2
11(a)	Order with respect to A: First order (rate proportional to [A] ¹) (1); Order with respect to B: Second order (rate proportional to [B] ²) (1)	2
11(b)	Rate = k[A][B] ² (1); Overall order = 1 + 2 = 3 (third order overall) (1)	2
12(a)	Comparing Exp 1 & 2: [S ₂ O ₈ ²⁻] doubles, rate doubles => First order in S ₂ O ₈ ²⁻ (1); Comparing Exp 1 & 3: [I ⁻] doubles, rate doubles => First order in I ⁻ (1)	2
12(b)	Rate = k[S ₂ O ₈ ²⁻][I ⁻] => k = Rate / ([S ₂ O ₈ ²⁻][I ⁻]) (1); k = (1.6 x 10 ⁻⁵) / (0.040 x 0.020) = 0.020 (1); Units: mol ⁻¹ dm ³ s ⁻¹ (1)	3
13(a)	Thiosulfate reacts rapidly with iodine as soon as it is formed, reducing it back to iodide (1); This keeps the solution colourless until all thiosulfate has been completely exhausted (1)	2
13(b)	Reaction 2 is extremely fast, so [I ₂] remains zero while any S ₂ O ₃ ²⁻ remains (1); The instant S ₂ O ₃ ²⁻ is completely consumed, newly formed I ₂ immediately complexes with starch, giving a sharp colour change (1)	2
14(a)	All mobile ions (Ba ²⁺ , OH ⁻ , H ⁺ , SO ₄ ²⁻) are removed from solution (1); forming insoluble solid BaSO ₄ and virtually un-ionised liquid water molecules (1)	2
14(b)	Conductivity increases again (1); Adding excess H ₂ SO ₄ introduces mobile H ⁺ and SO ₄ ²⁻ ions into the solution which carry electric current (1)	2
15(a)	Heterogeneous catalyst (1)	1
15(b)	Weigh the initial dry mass of MnO ₂ before reaction (1); Filter off the solid MnO ₂ after the reaction ceases, wash with distilled water and dry thoroughly (1); Reweigh the dried solid to show that its mass is unchanged (1)	3
16(a)	Initial rate = gradient of tangent at t = 0 = 44.0 / 40 = 1.10 cm ³ s ⁻¹ (2)	2
16(b)	Average rate = 48.0 / 60 = 0.80 cm ³ s ⁻¹ (1); As acid is consumed, [H ⁺] decreases, so collision frequency and instantaneous rate decrease continuously over the 60 seconds (1)	2
17(a)	Increases ONLY the collision frequency (fraction with E ≥ E _a remains unchanged at constant temperature) (1)	1
17(b)	Increasing temperature increases BOTH collision frequency and fraction with E ≥ E _a (1); The dramatic increase in rate is predominantly due to the much larger fraction of molecules having kinetic energy ≥ E _a (1); Collision frequency increase accounts for only a minor fraction (~2%) of the rate increase (1)	3
18(a)	Half-life (t _{1/2}) (1)	1
18(b)	First order (1); A constant half-life (t _{1/2} = 2.0 minutes) is the defining characteristic of a first-order reaction (1)	2
19(a)	Draw a tangent to the temperature-time curve at the start of the reaction (t = 0) (1); The initial rate of temperature rise (ΔT / Δt) is directly proportional to initial reaction rate (1)	2
19(b)	Heat loss lowers the recorded temperature rise below its true value (1); This reduces the measured gradient, underestimating the actual rate of reaction (1)	2
20(a)	Fine coal dust has an enormous total surface area to volume ratio compared to a solid lump (1); Tremendously higher frequency of fruitful collisions between oxygen molecules and carbon particles per second, releasing immense heat rapidly (1)	2
20(b)	Flour mills / grain silos (1); Flour / grain dust / starch particles (1)	2
21(a)	UV light photons provide the activation energy required for homolytic fission of Cl-Cl bonds (1); producing highly reactive chlorine free radicals (Cl·) that initiate a rapid chain reaction (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
21(b)	$\text{Cl}_2 \rightarrow 2\text{Cl}\cdot$ (under UV light / homolytic fission) (1)	1
22(a)	1-chlorobutane < 1-bromobutane < 1-iodobutane (1-iodobutane is fastest) (1)	1
22(b)	The rate-determining step involves breaking the carbon-halogen bond (C-X) (1); Bond enthalpy decreases down Group 17: C-Cl (346) > C-Br (290) > C-I (228 kJ mol ⁻¹) (1); The C-I bond is the weakest and has the lowest activation energy to break, leading to the fastest hydrolysis rate (1)	3
22(c)	Yellow precipitate of silver iodide, AgI (1)	1
23(a)	Continuous monitoring yields an unbroken, detailed real-time data curve without disturbing the reaction system (1)	1
23(b)	Reaction may continue slightly during the time required to withdraw and quench each sample (1); Human timing errors and titration end-point reading uncertainties occur for every sample (1)	2
24(a)	mol dm ⁻³ s ⁻¹ (1)	1
24(b)	$k = \text{Rate} / [\text{A}]^2 = (\text{mol dm}^{-3} \text{ s}^{-1}) / (\text{mol dm}^{-3})^2$ (1); Units = mol ⁻¹ dm ³ s ⁻¹ (or dm ³ mol ⁻¹ s ⁻¹) (1)	2
25(a)	Potassium has a larger atomic radius and more electron shielding, resulting in a lower first ionisation energy than sodium (1); Less energy is required to remove the valence electron, giving a lower activation energy and faster reaction rate (1)	2
25(b)	$2\text{K(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{KOH(aq)} + \text{H}_2\text{(g)}$ (2)	2
26(a)	No gas molecules have zero kinetic energy (particles are in constant motion) (1)	1
26(b)	The peak of the curve shifts down (lower height) and to the right (higher energy) (1); The curve broadens / flattens, with a higher tail extending into the high-energy region (1)	2
26(c)	The total number / amount of gas molecules in the sample (which remains constant) (1)	1
26(d)	At higher temperature T ₂ , a significantly larger fraction of molecules possess kinetic energy greater than or equal to the activation energy ($E \geq E_a$), represented by the shaded area (1); This causes a dramatic increase in the frequency of successful / fruitful collisions per unit time (1)	2
27(a)	A substance that increases the rate of a chemical reaction without being consumed / chemically changed at the end (1); by providing an alternative reaction pathway with lower activation energy (1)	2
27(b)	A catalyst lowers the activation energy from E_a to E_{cat} (1); A significantly larger fraction of molecules now possess kinetic energy $\geq E_{cat}$ (represented by the combined shaded areas), increasing successful collision frequency (1)	2
27(c)	The shape of the curve is completely unchanged (energy distribution depends only on temperature) (1)	1
28(a)	A catalyst that is in a different physical phase/state from the reactants (e.g. solid catalyst with gaseous reactants) (1)	1
28(b)	Stage 1: Reactant molecules diffuse to the surface and adsorb onto active catalytic sites (1); Stage 2: Interaction with metal surface weakens and stretches internal covalent bonds within reactant molecules, lowering E_a (1)	2
28(c)	Stage 3: Adsorbed fragments react together across the surface to form new product bonds (1); Stage 4: Product molecules desorb / detach from active sites and diffuse away, freeing sites for further reactants (1)	2
28(d)	If adsorption is too strong, product molecules cannot desorb or reactant bonds break irreversibly (1); Active sites become permanently blocked / poisoned, preventing further reaction (1)	2
29(a)	Collision frequency is proportional to the square root of absolute temperature ($\sqrt{T}$), which increases by only ~1.7% from 298 K to 308 K (1); However, the Maxwell-Boltzmann exponential distribution tail increases dramatically (1); The number of molecules with energy $\geq E_a$ roughly doubles, yielding twice as many fruitful collisions per second (1)	3
29(b)	Temperature rise = 50 - 20 = 30 °C = three 10 °C increments (1); Rate factor = 2 ³ = 8 times faster (1)	2
30(a)	(i) Heterogeneous (1); (ii) Homogeneous (1); (iii) Homogeneous (1); (iv) Heterogeneous (1)	4
31(a)	Both reactant ions ($\text{S}_2\text{O}_8^{2-}$ and I^-) are negatively charged (anions) (1); Intense electrostatic repulsion between like negative charges creates a high energy barrier for collision (1)	2
31(b)	Step 1: $\text{S}_2\text{O}_8^{2-} + 2\text{Fe}^{2+} \rightarrow 2\text{SO}_4^{2-} + 2\text{Fe}^{3+}$ (1); Step 2: $2\text{Fe}^{3+} + 2\text{I}^- \rightarrow 2\text{Fe}^{2+} + \text{I}_2$ (1)	2

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
31(c)	Each catalytic step involves collision between oppositely charged ions (cation and anion), which attract each other; Fe^{3+} simply carries out Step 2 first, regenerating Fe^{2+} (1)	1
32(a)	Step 1: $\text{SO}_2(\text{g}) + \text{NO}_2(\text{g}) \rightarrow \text{SO}_3(\text{g}) + \text{NO}(\text{g})$ (1); Step 2: $\text{NO}(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{NO}_2(\text{g})$ (1)	2
32(b)	NO_2 is regenerated chemically in Step 2 in equal amount and remains chemically unchanged overall (1)	1
33(a)	Provides an enormous surface area of active catalyst while minimising the amount of expensive precious metal required (1)	1
33(b)	$2\text{CO}(\text{g}) + 2\text{NO}(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{N}_2(\text{g})$ (2)	2
33(c)	Carbon monoxide and unburned hydrocarbons are oxidised to CO_2 , so the exhaust gas continues to contain carbon dioxide (a greenhouse gas) (1)	1
34(a)	Poison molecules adsorb strongly and irreversibly onto active catalytic sites on the solid surface (1); This physically blocks reactant molecules from adsorbing, rendering the active sites permanently inactive (1)	2
34(b)	Haber process: Sulfur / sulfur compounds (H_2S) or carbon monoxide (CO) (1); Catalytic converters: Lead (from leaded petrol) (1)	2
35(a)	A reaction in which one of the products acts as a catalyst for the reaction (1)	1
35(b)	Manganese(II) ions / $\text{Mn}^{2+}(\text{aq})$ (1)	1
35(c)	Rate starts very low (induction period) (1); Rate rises to a maximum as $[\text{Mn}^{2+}]$ builds up, then falls as reactants are consumed (bell-shaped rate curve) (1)	2
36(a)	Enzymes operate under mild physiological conditions (neutral pH, $\sim 37^\circ\text{C}$) (1); Enzymes exhibit extreme substrate specificity due to complementary 3D active site shape (1)	2
36(b)	Up to the optimum temperature, increasing temperature increases kinetic energy and fruitful collision frequency (1); Above optimum, thermal energy disrupts hydrogen bonds and tertiary folding of the protein (1); The enzyme denatures and the active site changes shape irreversibly, preventing substrate binding (1)	3
37(a)	(i) ΔH is unchanged (1); (ii) E_a of forward reaction is lowered (1); (iii) E_a of reverse reaction is lowered by the exact same amount (1)	3
37(b)	A catalyst only affects kinetics (reaction rate/pathway), not thermodynamics / free energy change / equilibrium position (1)	1
38(a)	The distribution is asymptotic to the x-axis; there is no theoretical upper limit to the kinetic energy a molecule can acquire through successive collisions (1)	1
38(b)	The curve is asymmetric with a long tail extending to very high energies on the right (1); These high-energy molecules pull the mean / average energy to a higher value than the mode / most probable energy ($E_{\text{mean}} > E_{\text{mp}}$) (1)	2
39(a)	Maximises the available surface area to volume ratio (1); Exposes vastly more active catalytic sites for adsorption of N_2 and H_2 molecules (1)	2
39(b)	The plant would need to operate at a much higher temperature (e.g. $> 700^\circ\text{C}$) to achieve an acceptable reaction rate (1); However, this would drastically reduce the equilibrium yield of ammonia to near zero (due to exothermic reaction) and require far more fuel (1)	2
40(a)	In V_2O_5 : +5 (1); In V_2O_4 : +4 (1)	2
40(b)	SO_2 acts as a reducing agent (reduces V from +5 to +4) (1); O_2 acts as an oxidising agent (re-oxidises V from +4 to +5) (1)	2
40(c)	Transition metals have partially filled d-orbitals and exhibit variable oxidation states (1); They can readily accept and donate electrons to form stable reaction intermediates (1)	2
41(a)	A catalyst increases the rates of forward and reverse reactions equally (1); It has zero effect on the position of equilibrium or yield of products (1)	2
41(b)	(i) Increases (1); (ii) Increases by the same factor (1); (iii) No change / unchanged (1)	3

Q	EXAMINER MARKING POINTS / INDICATIVE CONTENT	MARKS
42(a)	Physisorption involves weak intermolecular forces (London dispersion) with small ΔH (~ -20 to -40 kJ mol ⁻¹) (1); Chemisorption involves formation of covalent/coordinate chemical bonds with large ΔH (~ -100 to -400 kJ mol ⁻¹) (1)	2
42(b)	Chemisorption involves electron transfer or sharing that directly weakens internal reactant bonds (1); Weak physisorption does not weaken covalent bonds sufficiently to lower the activation energy (1)	2
43(a)	H ⁺ ions are dissolved in the same aqueous liquid phase as the reactants (1)	1
43(b)	Protonation adds positive charge to the carbonyl oxygen (1); This strongly withdraws electrons from the carbonyl carbon atom, making it much more electrophilic (δ^+) and vulnerable to attack by weak nucleophile H ₂ O (1)	2
44(a)	Partially filled 3d orbitals can accept electron pairs from reactant molecules into vacant orbitals (1); and donate electrons from filled d-orbitals to form temporary coordinate bonds and intermediate complexes (1)	2
44(b)	s-block metals have only one stable oxidation state (+1 for Na, +2 for Ca) and cannot participate in redox catalytic cycles (1); Transition metals readily undergo reversible one-electron and two-electron redox changes (1)	2
45(a)	No effect on reaction rate (1); Concentrations and collision frequency between NO and O ₂ molecules remain completely unchanged (1)	2
45(b)	Rate decreases (1); The volume must expand to maintain constant total pressure, which dilutes the reacting gases, lowering their concentrations and collision frequency (1)	2
46(a)	Heterogeneous catalysis (solid Ni with liquid oil and gaseous H ₂) (1)	1
46(b)	H-H covalent bonds undergo chemisorption and homolytic cleavage on the nickel surface (1); forming mobile, adsorbed hydrogen atoms on the metal lattice (1)	2
46(c)	Adsorbed intermediate can undergo reversible hydrogen atom loss before full hydrogenation (1); Rotation around the single bond before reforming the double bond can isomerise natural cis-double bonds into more stable trans-configurations (1)	2
47(a)	For high E_a , E_a lies further out on the exponential tail of the Maxwell-Boltzmann distribution (1); A 10 °C shift produces a proportionally far greater fractional increase in the shaded area $E \geq E_a$ compared to a low E_a barrier (1)	2
47(b)	Rate constant k increases exponentially with temperature (1)	1
48(a)	Step 1: Cl. + O ₃ → ClO. + O ₂ (1); Step 2: ClO. + O → Cl. + O ₂ (or ClO. + O ₃ → Cl. + 2O ₂) (1)	2
48(b)	Cl. and ClO. cancel out from both sides, leaving: O ₃ + O → 2O ₂ (or 2O ₃ → 3O ₂) (1)	1
48(c)	The chlorine radical (Cl.) is regenerated at the end of every cycle and continues the catalytic chain reaction repeatedly until chain termination occurs (1)	1
49(a)	Enormous geometric surface area for gas contact (1); Negligible back-pressure / does not impede exhaust gas flow and vehicle power (1)	2
49(b)	The catalytic converter requires high operating temperatures (around 300-400 °C) to reach its 'light-off' temperature (1); While cold, few exhaust gas molecules have kinetic energy $\geq E_a$ on the catalyst surface, so catalytic conversion is inefficient (1)	2
50(a)	The nitrogen molecule contains an exceptionally strong covalent triple bond (N≡N, bond energy = 945 kJ mol ⁻¹) (1); Tremendous kinetic energy is required to break this bond in homogeneous gas-phase collisions (1)	2
50(b)	N ₂ chemisorbs onto active iron surface sites where Fe d-orbitals back-donate electrons into N ₂ antibonding orbitals, cleaving the triple bond into adsorbed N atoms at much lower energy (1); Adsorbed N atoms react stepwise with adsorbed H atoms (1)	2
50(c)	Single distribution curve with E_{cat} (+160) positioned to the left of uncatalysed E_a (+335) (1); Shaded region under curve showing far greater fraction of molecules exceeding E_{cat} (1)	2
50(d)	At 450 °C with catalyst, reaction rate is commercially viable while preserving an acceptable equilibrium yield (since lower T favours exothermic forward yield) (1); Massive reduction in plant energy costs and less severe thermal wear on equipment (1)	2